IOT PROJECT

PROJECT TITLE : MULTIFUNCTION ROVER

TEAM MEMBER'S:

- DAGGULA SREE BHARGAV REDDY (Team lead)
- ARJUN ARUN NAMBIAR
- BASIREDDY SRAVAN KUMAR REDDY
- MALIGIREDDY ABHIRAMREDDY
- KODIDALA BABU

ABSTRACT :

Rovers are important for conducting in-situ scientific analysis of objectives that are separated by many meters to tens of kilometres. Current mobility designs are complex, using many wheels or legs. They are open to mechanical failure caused by the harsh environment. This thesis describes IOT Based Rover, a four wheeled rover capable of traversing rough terrain using an efficient high degree of mobility suspension system. The primary mechanical feature of the IOT Based Rover design is its drive train simplicity, which is accomplished by using only two motors for mobility. Both motors are located inside the body where thermal variation is kept to a minimum, increasing reliability and efficiency. Four wheels are used because there are few obstacles on natural terrain that require both front wheels of the rover to climb simultaneously. This can be done in 2WD,4WD.6WD according to the need of the purpose.

APPLICATIONS:

- Remote areas can also be explored.

COMPONENTS:

- 2 (or) 4 (or) 6 x gear motor
- 2 x tires (or) 4 x tires (or) 6 x tires

- 2 (or) 4 (or) 6 x motor fixed pieces
- 2 x 100 * 213 * 5 mm plate
- 1 x L298N motor driver board
- 1 x Uno R3 board
- 1 x holder
- 1 x SG90 servo
- 1 x ultrasonic sensor module
- 2 x 18650 Battery pack
- 1 x HC-06 Bluetooth module
- 1 X40Pin DuPont line
- 1 x USB cable
- 1 x set of screws
- 1 x 6v solar panel
- 1 x TP4056 Charging Module – Type C
- 4 x 18650 li-ion Battery 3.7v
- 2 Pin Switch
- 5 mm Led
- 1K Resistors